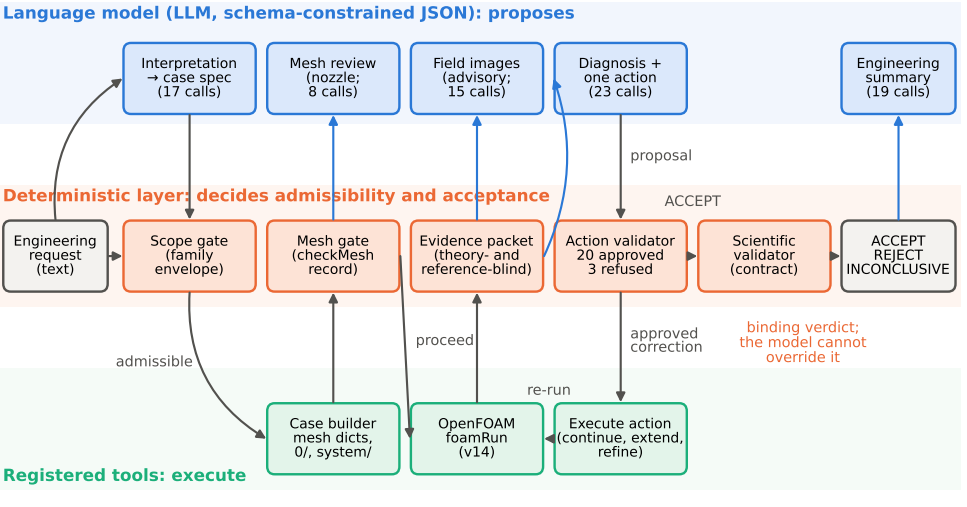

Language model (LLM, schema-constrained JSON): proposes
Interpretation → case spec (17 calls)
Mesh review (nozzle; 8 calls)
Field images (advisory; 15 calls)
Diagnosis + one action (23 calls)
Engineering summary (19 calls)
proposal
Deterministic layer: decides admissibility and acceptance
ACCEPT
Engineering request (text)
Scope gate (family envelope)
Mesh gate (checkMesh record)
Evidence packet (theory- and reference-blind)
Action validator 20 approved 3 refused
Scientific validator (contract)
ACCEPT REJECT INCONCLUSIVE
admissible
proceed
approved correction
binding verdict; the model cannot override it
re-run
Case builder mesh dicts, 0/, system/
OpenFOAM foamRun (v14)
Execute action (continue, extend, refine)
Registered tools: execute